Topic C5: Monitoring and controlling chemical reactions

C5.1 Monitoring chemical reactions

Summary
This topic tackles the relationship of moles to the concentration of a solution and the volume of a gas. It also tackles the calculation of the mass of a substance in terms of its molarity. The topic then moves on to look at using equations to make predictions about yield by calculations and to calculate atom economy.

Underlying knowledge and understanding
Learners should be familiar with the mole from Topic C3 and know that it measures the amount of substance. They should be familiar with representing chemical reactions using formulae and using equations.

Common misconceptions
The most common problem learners' encounter with these calculations is their lac of understanding of ratios. Also most learners think that the mole and mass are th same thing. This is reinforced by use of phrases such as '1 mole is 12 g of carbon, '1 mole is the relative atomic mass in grammes' or '1 mol = 12 g C' in teaching anc in textbooks equating amount of substance to mass, portion of substance, numbe of particles (Avogadro's number) or number of moles. All these phrases reinforce the idea that amount of substance is a measure of mass or a number.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM5.1i ☑	**calculations with numbers written in standard form when using the Avogadro constant**	**M1b**
CM5.1ii ☑	provide answers to an appropriate number of significant figures	M2a
CM5.1iii ☑	convert units where appropriate **particularly from mass to moles**	M1c
CM5.1iv ☑	arithmetic computation, ratio, percentage and multistep calculations permeates quantitative chemistry	M1a, M1c, M1d
CM5.1v ☑	arithmetic computation when calculating yields and atom economy	M1a, M1c
CM5.1vi ☑	change the subject of a mathematical equation	M3b, M3c

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
C5.1a ☑	**explain how the concentration of a solution in mol/dm^3 is related to the mass of the solute and the volume of the solution**		M1b	WS1.3c, WS1.4a, WS1.4c	Making standard solutions.
C5.1b ☑	describe the technique of titration				Acid/alkali titrations. (PAG C6)
C5.1c ☑	**explain the relationship between the volume of a solution of known concentration of a substance and the volume or concentration of another substance that react completely together**	**titration calculations**	M2a, M1c	WS1.3c, WS1.4a, WS1.4b, WS1.4c	
C5.1d ☑	**describe the relationship between molar amounts of gases and their volumes and vice versa**		M1c	WS1.3c, WS1.4a, WS1.4c, WS1.4d, WS1.4f	Measurement of gas volumes and calculating amount in moles. (PAG C8)
C5.1e ☑	**calculate the volumes of gases involved in reactions using the molar gas volume at room temperature and pressure (assumed to be 24 dm^3)**		M1b, M1c		
C5.1f (Combined Science C3.1j)	**explain how the mass of a solute and the volume of the solution is related to the concentration of the solution**		M1b, M1c	WS1.3c, WS1.4a, WS1.4c	
C5.1g ☑	calculate the theoretical mass of a product from a given mass of reactant.		M1a, M1c, M1d	WS1.3c	

C5.1h ☑	calculate the percentage yield of a reaction product from the actual yield of a reaction		M1a, M1c, M1d	WS1.2a, WS1.2b, WS1.2c, WS1.2d, WS1.3c, WS2a, WS2b	
C5.1i ☑	define the atom economy of a reaction				
C5.1j ☑	calculate the atom economy of a reaction to form a desired product from the balanced equation		M1a, M1c	WS1.3c	
C5.1k ☑	**explain why a particular reaction pathway is chosen to produce a specified product given appropriate data**	**data such as atom economy (if not calculated), yield, rate, equilibrium position and usefulness of by-products**	M3b, M3c	WS1.3c, WS1.3f	